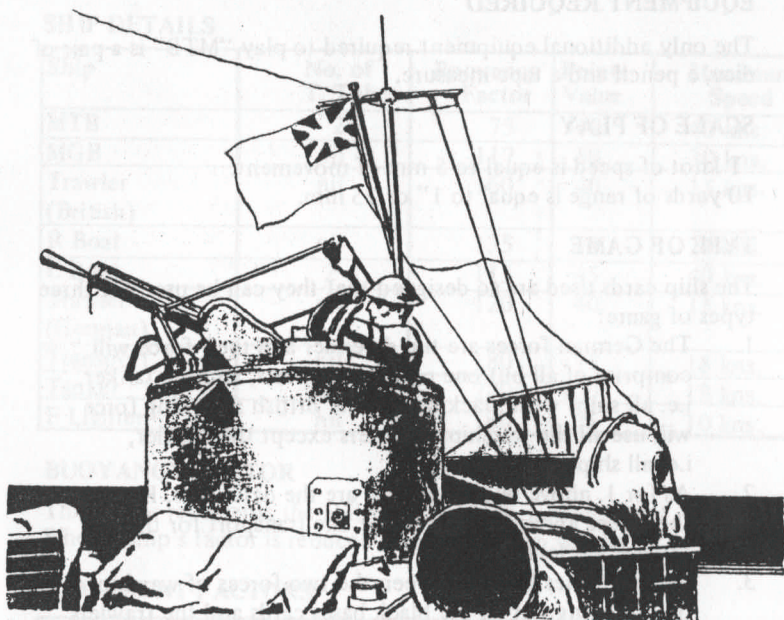


MTB



A TABLETOP GAME FOR ADULTS AND CHILDREN



SITUATION

The time 1942/43, the place is the English Channel. The scene set is that of the almost continual battle for supremacy between the small boat navies of Germany and Great Britain. The prize for both sides is two-fold: first, the elimination of the other's coastal convoys, and second, the control of this narrow and dangerous stretch of water.

The majority of ships involved were, for the Germans, R and E boats, and, for the British, MGBs and MTBs. Both sides also pressed into service, as convoy escorts and mine sweepers, trawlers from their fishing fleets.

The game "MTB" re-creates the main aspect of this battle: that of the attack and defence of coastal convoys. You control a small but powerful force of these craft and your orders are, as the attacker, to destroy as many enemy ships as you are able in one attack and, as the defender, to beat off the attack and get your convoy through.

EQUIPMENT REQUIRED

The only additional equipment required to play "MTB" is a pair of dice, a pencil and a tape measure.

SCALE OF PLAY

1 knot of speed is equal to 5 mm of movement.
10 yards of range is equal to 1" or 25 mm.

TYPE OF GAME

The ship cards used are so designed that they can be used for three types of game:

1. The German forces are the defender and their force will comprise of all but one of the transports, and the tanker, i.e. all ships with black bases. The British attacking force will use all the warships counters except the trawler, i.e. all ships on red bases.
2. As for 1. above but the British are the defenders. Reverse the forces above but substitute one transport for the three F lighters.
3. An encounter battle between the two forces of warships only, leaving out all the black bases cards and the trawlers on each side.
(NOTE: In this game a dice must be thrown to determine which side is to be classed as the attacker.)

SHIPS USED

The forces used for each side are:

BRITISH	GERMAN	THE CONVOY
4 MTBs	5 E Boats	1 Tanker
3 MGBs	2 R Boats	3 Transports
1 Trawler	1 Trawler	3 F Lighters



SPEED

Ships may move at maximum speed throughout the game, but a ship moving at less than its maximum speed may not increase its speed by more than 20 knots per move. A ship slowing down may not decrease its speed by more than 10 knots in any move unless the speed decrease is due to damage.

SHIP DETAILS

Ship	No. of T. Tubes	Buoyancy Factor	Points Value	Maximum Speed
MTB	2	75	40	40 kns.
MGB	nil	117	50	30 kns.
Trawler (British)	nil	160	40	15 kns.
R Boat	nil	135	32	24 kns.
E Boat	2	115	45	40 kns.
Trawler (German)	nil	120	40	15 kns.
Transport	nil	180	75	18 kns.
Tanker	nil	270	120	18 kns.
F Lighter	nil	80	25	10 kns.

BUOYANCY FACTOR

This factor represents the ability of a ship to remain afloat. When a ship's factor is reduced to zero the ship will sink.

ARMAMENT FACTORS

Ship	No. of depth charges	Weapon Factors		
		Broadside	Forward	Astern
*MTB	nil	30	20	40
*MGB	22	100	80	80
*Trawler (British)	nil	40	30	10
*R Boat	4	60	40	50
E Boat	nil	60	40	50
*Trawler (German)	6	40	30	10
*Transport	nil	20	20†	20†
*Tanker	nil	20	20	nil
*F Lighter	nil	50	70	40

Only ships marked * may use half their broadside or forward weapon factor at targets that are over 500 yards range.

Ships marked † have weapons disposed in different positions. Check ship card to determine where.

SETTING UP THE GAME

Throw one dice per player taking part; the highest dice taking the attacker's forces, the lowest taking the defender's. Set up the defending force in the middle of the playing area and at one end. The attacking force is then set up anywhere within 10" of the other end of the playing area. The game commences when each side has written its first set of orders.

DECIDING THE WINNER

At the end of the game, the duration of which is ten moves, the winner will be determined as follows:

ATTACKER

- 1 per 10 buoyancy points lost.
- + 1 per 10 buoyancy points lost by defender's escorting warships.
- plus the points value of any enemy ships sunk.

DEFENDER

- 1 per 10 buoyancy points lost by escorts
- plus points value of any enemy ships sunk
- plus points value of all merchant ships still afloat

(NOTE: Any ships on fire at the end of the game will not count in any of the above calculations.)

If the attacker's score is higher than the defender's the attacker is the victor. If the scores are equal or the defender's score is the higher the defender is the victor.

WRITING ORDERS

Movement orders must be written as follows:

4P ST 30K

4P...Indicates the ship will make a 4 point turn

ST...Indicates the ship will turn starboard

30K...Indicates the ship will be moving at 30 knots.

(NOTE: If a player wishes to make his turn at the end of the move the order is written as: 30K 4P ST.)

COMPULSARY ORDERS

If, due to damage, a ship is ordered to turn away, write 'TA' before the move order. TA indicates that the ship must turn away from the enemy and withdraw towards its own base line at the best possible speed.

If the ship catches fire write 'OF' before the move order.



MOVEMENT STAGES

Each side will take it in turns to move first. The attacker will move first on all ODD numbered moves and the defender on all EVEN numbered moves. The player who moves first will be called the phasing player.

The following procedure should be followed each move, after both sides have completed their order writing.

- | | |
|----------|--|
| Stage 1 | Declare any torpedo attacks and place a torpedo marker at the stern of the firing ship. The firer must state at this stage which enemy ship is the target. |
| Stage 2 | Move phasing player's ship. |
| Stage 3 | Resolve any depth charge attacks. |
| Stage 4 | Move non-phasing player's ships. |
| Stage 5 | Resolve phasing player's fire. |
| Stage 6 | Resolve non-phasing player's fire. |
| Stage 7 | Resolve Torpedo attacks. |
| Stage 8 | Resolve results of any ramming. |
| Stage 9 | Adjust all ship's factors resulting from fire and other attacks, and determine any compulsory actions. |
| Stage 10 | Write orders for the next move. |

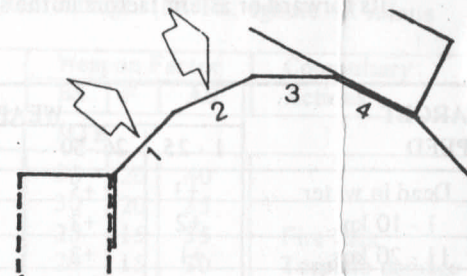
TURNING

Each point turn that a ship makes will be the equivalent of three knots in speed.

A ship may only make a turn at the beginning of the move or at the end, and at no other time.

Turns are measured by using the turning fans supplied: lay the straight edge of the fan against the side of the ship counter, then move the ship the required number of points, keeping the fan stationary.

The following is an example of a ship making a 4 point turn to starboard. The total distance moved during the turn is 12 knots.



RANGES

Measure all ranges from the center of the firing ship counter's edge, ie. the sides of the counter for broadsides, the ends for forward or astern fire, to the nearest point on the target counter.

ARCS OF FIRE

A ship has three arcs of fire: **BROADSIDE** (port or starboard), **FORWARD** or **ASTERN**.

A ship may only engage a target which is completely within its arc of fire, ie. all the target counter is within the engaging arc. At ranges of 25 yards or less only half the target counter need be in the engaging arc.

To measure an arc of fire lay the turning fan base alongside the ship counter edge and check, with a tape measure, that all the target counter is within the appropriate arc on the fan. If it is not the ship may not engage with that weapon factor.

OVERHEAD FIRING

Boats may not fire over each other unless the firing ship is a transport or a tanker and the target is not within 10" of an intervening ship.

Transport and tankers may not fire over each other at any time, nor may any ship fire over one which is on fire.

FIRING

To determine the effect of one ship firing at another first determine the weapon factor that can bear, then determine the range to the target and also the target's speed. Using those details and the factors from the following tables determine the total damage points caused. Each damage point will be equal to one buoyancy point lost. (See later tables for the effect.)

(NOTE: A ship may only use its forward or astern weapon factors if it is not using its broadside weapon factor that move. A broadside weapon factor may only be used once per move, either to port or starboard, but not both. A boat may not use its forward or astern factors in the same move.)

TARGET SPEED	WEAPON FACTOR			
	1 - 25	26 - 50	51 - 75	76 - 100
Dead in water	+3	+5	+8	+10
1 - 10 kns.	+2	+3	+5	+ 8
11 - 20 kns.	+1	+2	+3	+ 5
21 - 30 kns.	0	0	+1	+ 3
Over 30 kns.	0	0	0	+ 1

TARGET

SIZE	WEAPON FACTOR	
	1 - 50	51 - 100
Target		
MTB	- 3	- 2
E Boat	- 2	- 2
R Boat or MGB	- 1	+1
Trawler	0	+3
Lighter	+1	+2
Transport	+2	+4
Tanker	+3	+5

+3 if engaged the same target with the same weapon factor last move.

RANGE TO TARGET

RANGE IN YARDS	FACTOR
0 - 25	+5
26 - 50	+4
51 - 100	+3
101 - 200	+1
201 - 300	0
301 - 400	- 1
401 - 500	- 2
Over 500	- 4

RANDOM EFFECT

DICE THROW	WEAPON FACTOR			
	1 - 25	26 - 50	51 - 75	76 - 100
1	- 4	- 3	- 2	- 1
2	- 2	- 1	0	+ 1
3	0	0	+3	+ 4
4	+2	+3	+5	+ 7
5	+3	+7	+7	+ 8
6	+5	+6	+8	+10

RESULTS OF FIRING

After adding all the previous factors reduce the boat's buoyancy factor by that amount, then consult the following tables to determine the result of the damage. (NOTE: Ignore all minus results.)

Buoyancy Factor	Speed	Weapon Factor			Compulsary Actions
		BS	F	AS	
MTB					
75 - 60	1 dice	30	20	40	Fire Risk Torpedo damage Fire Risk Fire Risk/turn away Dead in water
59 - 50	1 dice	30	20	35	
49 - 40	1 dice	25	15	35	
39 - 30	2 dice	20	15	30	
29 - 20	3 dice	15	10	25	
10 - 10	3 dice	10	5	15	
9 - 1	-	5	0	10	

Buoyancy Factor	Speed	Weapon Factor			Compulsary Actions
		BS	F	AS	
MGB					
117 - 90		100	80	80	Fire Risk
89 - 70	1 dice	95	75	75	
69 - 50	1 dice	85	65	75	
49 - 30	2 dice	75	60	65	
29 - 20	2 dice	65	55	60	Depth charge damage
19 - 15	3 dice	45	35	40	Fire Risk/turn away
14 - 1		30	20	30	Dead in water
TRAWLER					
160 - 100		40	30	10	Fire Risk
99 - 75		35	25	10	
74 - 50	1 dice	30	20	10	
49 - 30		25	15	10	
29 - 15	2 dice	15	15	0	Fire Risk
14 - 1		10	10	0	Dead in water
'E' BOATS AND 'R' BOATS					
135 - 100		60	40	50	Fire Risk Torp./d. charge damage
99 - 80	1 dice	55	35	50	
79 - 60	1 dice	50	25	40	
59 - 40	2 dice	45	25	35	
39 - 30	2 dice	35	15	30	Fire Risk Turn away Dead in water
29 - 20	3 dice	25	15	20	
19 - 15	3 dice	25	10	15	
14 - 1		15	10	10	
TRANSPORTS					
180 - 100		20	20	20	Fire Risk
99 - 50	1 dice	15	15	15	
49 - 25		10	10	10	
24 - 10	2 dice	5	5	5	
9 - 1		0	0	0	Dead in water
TANKER					
270 - 200		20	20	0	Fire Risk Fire Risk Fire Risk Dead in water
199 - 100	1 dice	15	15	0	
99 - 50		10	10	0	
49 - 25	2 dice	5	5	0	
24 - 11		0	0	0	
'F' LIGHTER					
80 - 50		50	70	40	Fire Risk
49 - 25		35	60	40	
24 - 15	1 dice	35	55	25	
14 - 10		30	45	20	
9 - 1		20	30	10	Dead in water

SPEED REDUCTION

Where 1 or 2 dice are indicated in the 'Speed' column of the previous table throw either one or two dice. The result will be the number of knots the ship will lost next move.

DAMAGE TO TORPEDO AND DEPTH CHARGE EQUIPMENT

Where indicated on the 'Compulsary Actions' column throw one dice and consult the table below.

DICE THROW	TORPEDOES	DEPTH CHARGES
1 or 2	Fire control knocked out	Port discharged jammed.
3	Starboard tube knocked out.	Starboard discharger jammed
4	Port tube knocked out.	No damage.
5 or 6	No damage.	

FIRE RISK

As damage is sustained by a boat there is a risk of fire breaking out. Where 'Fire Risk' is indicated in the 'Compulsary Action' column throw two dice and add or subtract the following:

- + 1 If ship is a transport or trawler
- + 2 If ship is an F lighter
- 1 If boat is an R boat
- 2 If boat is an MTB, MGB, or E boat
- 3 If ship is an tanker

RESULT: If the final total is 5 or less fire will break out. Once the fire has started the crew of the vessel have three moves in which to get it under control, before the vessel blows up.

Any vessel within 50 yards of another vessel, which blows up, will sustain damage, the amount of damage being determined by throwing one dice and multiplying the result by 10. The result will be equal to the amount of buoyancy points lost by that vessel.

A vessel which blows up whilst locked to another after a ram*, will cause the other to blow up with it.

(*See ramming rules later.)

PUTTING OUT A FIRE

To get a fire under control throw two dice at the end of each subsequent move and consult the following:

- First move..... 8 to 12 required
- Second move..... 10 to 12 required
- Third move..... 11 or 12 required.

(NOTE: Vessels on fire may not use any of their weapon.)

DEPTH CHARGE ATTACKS

This form of attack was used against surface ships on a few occasions. However, it proved as hazardous to the attacker as to the target ship. The idea was to move at maximum speed at the stern of the target vessel, throwing the wheel over at the last moment and dropping two depth charges under the target vessel's stern. The charges were set at minimum depth and, if they exploded near enough, they would blow off the target's screws or damage its rudder. As a partial safety measure the charges could be set to go off deeper, causing less damage.

A player must state whether the charges are set to deep or shallow when ordering this form of attack.

METHOD OF ATTACK

Move the attacking vessel through its full move stage, leaving a depth charge marker at the nearest point to the target's stern. Throw one dice* per attacking vessel and one dice for the target vessel, adding or subtracting the following to determine the damage caused to those involved.

ATTACKER

- 1 Per 1", or part of, away from vessel's stern.
- +2 Per charge dropped.
- +2 If speed less than 10 kns.
- 1 If speed 10 to 20 kns.
- 1 If speed over 30 kns.
- 2 If charges set deep.
- +1 If charges set shallow

DEFENDER

- 1 Per 1", or part of away from vessel's stern.
- +1 Per charge dropped.
- 1 If speed over 30 kns.
- +1 If speed 20 to 30 kns.
- +2 If speed 10 to 19 kns.
- +3 If speed less than 10 kns.
- 2 If charges set deep.
- +2 If charges set shallow.

RESULT

	ATTACKER	DEFENDER
1 or less	No damage	10 buoyancy points lost.
2	10 buoyancy points lost.	15 buoyancy points lost.
3	15 buoyancy points lost.	25 buoyancy points lost.
4	25 buoyancy points lost.	Rudder jammed for 2 moves, 25 buoyancy points lost.
5 or more	Rudder jammed for 2 moves, 30 buoyancy points lost.	Screws blown off, rudder jammed, 50 buoyancy points lost.

*Dice throw: count 1 or 2 as -1
count 3 or 4 as 1
count 5 or 6 as 2

TORPEDO ATTACKS

A ship may only fire torpedoes if its fire equipment is undamaged and its buoyancy factor is not less than a quarter of its starting buoyancy factor. Torpedo firing orders must be written in the move orders, and must include the target at which they are to be filed.

Mark the position from which the torpedoes are to be fired by placing a torpedo marker at the stern of the firing vessel before that vessel starts its movement.

At the start of stage 7 lay a tape measure from the torpedo marker to the target vessel, and determine if there are any intervening ships. Measure the range. If there are intervening ships consult the list below; if not calculate the effect of the torpedoes.

INTERVENING SHIPS

1. If a warship, other than a trawler, travelling at 20 knots or over, the torpedoes will pass under it.
2. If a transport or trawler, the torpedoes will hit automatically and no calculations are needed.
3. If an F Lighter, the torpedoes will pass under it, unless the target was another F Lighter, in which case the hit will be automatic.
4. If a warship, other than a trawler, travelling at less than 20 knots, the torpedoes will hit automatically only if the original target was NOT a tanker.

TORPEDO CALCULATIONS

If there is no intervening ship, throw two dice and add or subtract the following:

- 1 Per 100 yards, or part of, range to target.
- 1 If target turns more than 2 points during the move.
- 1 If target speed is more than 20 and less than 31 knots.
- 1 If target is smaller than a transport.
- 2 If target speed is more than 30 knots.
- +2 If target speed is less than 20 and more than 10 knots.
- +3 If target speed is 10 knots or less.
- +5 If target is dead in the water.
- +2 If target is a tanker.
- +1 Per Torpedo fired.
- +1 If target is a transport.



RESULT

A final score of 7 or more will result in one torpedo hit, 10 or more two hits if both torpedoes fired at the same time and the same target.

EFFECT OF A TORPEDO HIT

Throw one dice for the first hit and consult the table below. Two hits, at the same time will automatically blow up the target. A hit against any vessel smaller than a trawler will automatically sink it.

DICE THROW	RESULT
1 to 4	Vessel sinks next move.
5 or 6	Vessel loses 100 buoyancy points.



RAMMING

If, at the end of a move, one ship counter finished on top of, or touching another ship counter the former is considered to have rammed the latter. There are two types of ram:

ACCIDENTAL, where the ramming vessel was NOT ordered to ram.

INTENTIONAL, where the ramming vessel WAS ordered to ram. After a ram has occurred the following will apply to those vessels involved.

1. A warship, rammed by a transport or tanker, will end the move travelling in the same direction. The transport or tanker will make a one point turn away from the rammed ship.
2. A warship, rammed by another warship, trawler or lighter, will end the move stationary, and both will lock together. They will remain locked together for the next move, during which neither may use its weapons. In the following move they will disengage and may proceed and use weapons as ordered.
3. A warship ramming a transport or tanker will have the following result:
Transport will turn 2 points away from the warship.
Tanker will turn 1 point away from the warship.
Warship will turn and end the move travelling in the same direction as the rammed vessel.

EFFECTS OF A RAM

After a ram has occurred determine the damage to each ship in the following way:

First, consult the table below and determine the ramming factor for each vessel involved.

FACTORS:

Rammed Ship	Ramming Ship
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RAMMED SHIP	RAMMING SHIP				
	MTB	R BOAT E BOAT	MGB	TRAWLER LIGHTER	TRANSPT. TANKER
TANKER	12:30	15:28	18:28	15:26	24:28
TRANSPORT	15:25	18:25	20:24	18:22	32:30
TRAWLER	18:20	20:22	24:22	20:18	35:25
MGB	20:16	24:18	26:18	22:15	38:20
R BOAT	23:15	25:15	30:14	24:13	42:15
E BOAT	25:15	25:14	30:14	25:12	45:12
MTB	30:15	28:10	32:10	30:10	50:10
LIGHTER	15:14	16:12	18:10	20:10	40:8

Next, throw two dice per ship and subtract the total from each ship's ramming factor found above.

Finally, add or subtract the factors from the following list:

RAMMED SHIP

- +10 If intentional ram.
- + 5 If accidental ram.
- + 1 Per knot of speed of ramming ship, over 10 knots.

+10 If dead in the water.

- + 8 If broadside ram.
- + 4 If stern ram.

RAMMING SHIP

- + 5 If intentional ram.
- + 3 If accidental ram.
- 2 Per knot of speed of rammed ship over 10 knots.
- + 5 If rammed ship dead in the water.
- + 8 If broadside ram.
- 5 If stern ram.

(NOTE: Head on ramming will not occur. It is assumed that ships will have time to avoid one another.)

The total result of each ship will be the number of buoyancy points lost.

(NOTE: Vessels involved in a ram will still be able to use their weapons in the move during which the ramming occurs.)

OPTIONAL GAMES

The following rules apply when players wish to introduce variations to visibility, ie. night, fog or misty conditions.

VISIBILITY

Throw one dice and consult the following:

1 or 2 Misty conditions.

3 or 4 Fog patches.

5 or 6 Action will take place at night.

MISTY CONDITIONS AND FOG

For misty conditions and fog patches proceed as for the normal game. Each player will place onto the playing area one 18" square of paper, representing the position of a fog or mist bank. Visibility in fog patches will be 25 yards, and ships outside them will be unable to see into the mist patches unless they are within 150 yards of a patch.

FIRING

Having determined the type of visibility use the table below in conjunction with the Fire Effect tables on page 7.

VISIBILITY	RANGE IN YARDS			
	0-100	101-250	251-500	Over 500
Mist	- 1*	-2	-3	-5
Night-no illumination	- 2*	-4	-7	-9
Night-with illumination	+1	0	-2	-3

* Ignore if range is 25 yards or less, and the MOONLIGHT factor in night actions is 3 or more.



NIGHT ACTIONS MOONLIGHT FACTOR

Throw one dice at the start of each odd numbered move. The score will be equal to the degree of moonlight during the next two moves.

ADDITIONAL EQUIPMENT REQUIRED

Three or four squares of card, 12" square with a 9" diameter circle cut out in the center of each.

One dummy counter, 1" wide by 2" long, for each vessel taking part. Each counter should be numbered for identification.

SETTING UP THE GAME

Set up the game as detailed in the normal game but substitute the dummy counters for the ship counters.

SIGHTING

Each move the automatic sighting distance will be 3 times the moonlight factor in inches. Within this distance ships will automatically sight each other.

To sight a vessel outside this distance, measure the range from the sighting ship to the target. Throw two dice and add or subtract the following factors:

(NOTE: Only two attempts per side per move may be made.

Targets may only be engaged if they have been sighted.)

-1 Per 100 yards range, or part of.

-1 If target is travelling at 18 to 29 knots.

-2 If target is travelling at 10 to 17 knots.

-2 If sighting ship is travelling at over 15 knots.

-4 If target is stationary or travelling at 10 knots or less.

+1 Per moonlight factor.

+1 If target is travelling at 30 knots or more.

+5 If target fired during the move.

Sighting and identification of the target will take place if the final score is 6 or more.

Once sighting has taken place substitute the actual ship counter for the dummy one. The target may be engaged by the sighting ship only, in the next move. It may be engaged by other ships on the following move without those ships having sighted it.

Dice for possible sightings at the end of each move before writing orders for the next move.

ILLUMINATION

Illumination will be by rocket projector or by star shell, each of which will illuminate a 9" diameter area for two moves. Two projectors will be carried by each warship, excluding the trawler, and each will be capable of firing once per game.

Star shells may be fired by a trawler or an F lighter in a German convoy, and by a trawler or a transport in a British convoy. These ships may fire star shells each move. A vessel firing illumination rounds may not use its broadside weapon factor in the same move. Once a target has been illuminated it may be engaged by any enemy vessel without prior sighting calculations.

Once a target has been sighted it may be illuminated at the beginning of the next move, at Stage 1.

Place the center of a 9" diameter circle on the center of the target vessel; all other vessels wholly or partly covered by the circle will also be identified.

VESSELS OUTSIDE THE ILLUMINATED AREA

When a target vessel is no longer illuminated, and is outside the automatic sighting range, remove the ship counter and replace it with a dummy counter.

If the vessel is within automatic sighting range, or if it is being engaged by the same ship which was engaging it while it was illuminated, do not remove the ship counter. An engaging ship may continue to fire at a target even if it is no longer illuminated and is outside the automatic sighting range. Once the engaging ship ceases firing substitute the dummy counter.

TORPEDO FIRING

Torpedoes may only be fired at a target which was illuminated or sighted last move.

In calculating torpedo hits use the factors below in conjunction with the 'TORPEDO CALCULATIONS' on page 12.

- 1 If firing in misty conditons or at night without illumination.
- 5 If firing in fog conditions, and target covered by fog bank.
- +1 If firing at night and the moonlight factor is greater than 3.



COMBAT ORDER CHART

SHIP	INITIAL WEAPONS ETC		MOVE																				WEAPON FACTORS REMAINING	
			1	2	3	4	5	6	7	8	9	10												
	TT	DC																					F	
	BF	MS																					R	
			BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	A	
	TT	DC																					F	
	BF	MS																					R	
			BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	A	
	TT	DC																					F	
	BF	MS																					R	
			BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	A	
	TT	DC																					F	
	BF	MS																					R	
			BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	A	
	TT	DC																					F	
	BF	MS																					R	
			BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	BF	MS	A	

TT Torpedo Tubes - DC Number of Depth Charges BF Buoyancy Factors MS Maximum Speed F Forward B Broadside A Astern weapon factors

